



Project Management Skills-20PM01T _ Notes REV-01-2025

Unit 01: Introduction

Topics to be covered:

- 1.1 Introduction
- 1.2 Meaning of Project
- 1.3 Definition of Project
- 1.4 Features of a Project
- 1.5 Types of Projects
- 1.6 Benefits of Project Management
- 1.7 Obstacles in Project Management
- 1.8 Project Manager and His Role
- 1.9 Project Consultants
- 1.10 Meaning of Operation
- 1.11 Difference between Project and Operation
- 1.13 Process in Project Management
- 1.14 Project Scope and project objectives



1.1 Introduction:

Project management skills help in planning, organizing, and completing projects successfully. These skills include managing time, resources, and risks while working with a team to meet goals.

Good project management ensures tasks are completed on time and within budget. Learning these skills is important for handling challenges and achieving success in any project.

1.2 Meaning of Project:

The meaning of a project refers to a temporary and unique effort undertaken to achieve specific goals or objectives within a defined timeframe, scope and resources.

1.3 Definition of Project

- ☐ A project is a temporary activity, unique task that must be completed within a fixed budget and has a definite start and end date.

OR

- ☐ Project is a temporary activity undertaken to create a unique product or service.

Examples for project:

- ☐ Building a house (Construction projects).
- ☐ Railway projects.
- ☐ Metro rail project.
- ☐ Highway Projects.
- ☐ Finding a new vaccine.
- ☐ IT projects.
- ☐ Developing a new app.
- ☐ Developing new software.
- ☐ Health projects.
- ☐ Research projects.
- ☐ Educational projects.
- ☐ Power projects.
- ☐ Oil refinery projects.
- ☐ Conducting national elections.
- ☐ War execution.
- ☐ Launching new weapon.
- ☐ Automobile projects.
- ☐ Manufacturing projects.

1.4 Features of a Project

- ☐ Unique in nature. (No two projects are exactly similar)
- ☐ Clear Scope (The work and goals are well-defined).
- ☐ Creates Change (Projects aim to bring something new or better).
- ☐ Require set of resources. (Time, money, materials, machines, man power etc.)
- ☐ Have a fixed time to complete.
- ☐ Temporary (Projects have a clear start and end date).
- ☐ Connected Tasks (Projects involve related activities).
- ☐ Involves risk and uncertainty.
- ☐ Project has a life cycle reflected by start, growth, maturity and decline
- ☐ Require cross-functional teams and interdisciplinary approach.
- ☐ Change is an inherent feature in any project throughout its life.

1.5 Types of Projects

Projects can be classified as follows:

- i. Normal project
- ii. Crash project
- iii. Disaster project

i. Normal project:

- ☐ A project completed within the planned time, budget and resources without major changes or urgency.
- ☐ All the stages in this project are allowed to take their normal time as planned previously.
- ☐ Minimum capital is required. (Money or Fund)
- ☐ No sacrifice in quality.
- ☐ Example: Constructing a small house within the planned cost and time.

ii. Crash project:

- ☐ A project where extra resources or effort are used to finish it faster, usually at a higher cost.
- ☐ Requires additional cost to gain time.
- ☐ Simultaneous work by sub-contracting.
- ☐ Minimum overlapping of phases (stages) is encouraged.
- ☐ Ex: construction of building, construction of highways etc.

iii. Disaster project:

- ☐ A project that faces big problems, like poor planning, mistakes, or unexpected events, leading to failure or major delays and losses.
- ☐ Natural calamities such as flood, earthquake etc. are also few examples for unexpected events.
- ☐ Anything can be needed to gain time (huge or massive investments).
- ☐ Round the clock work is done.
- ☐ Example 1: Building a bridge that collapses due to poor planning, resulting in delays and financial loss.
- ☐ Example 2: Repair of dam due to faulty materials, resulting in delays and financial loss.
- ☐ Example 3: Repair of bridges or roads in case of flood or any other disaster.

1.6 Benefits of Project Management

Here are some benefits of project management:

- ☐ **Clear Goals:** Helps to set and achieve specific targets.
- ☐ **Good Planning:** Helps to manage tasks, time and resources.
- ☐ **Avoid losses:** Avoids wasting of time, money and materials.
- ☐ **Risk Management:** Find and solve problems early.
- ☐ **Time management:** Helps to complete the project on time.

- ☐ **Cost Management:** Helps to complete the project within budget.
- ☐ **Proper Communication:** Keeps the team and stakeholders connected.
- ☐ **Quality Management:** Complete the work with required standards.
- ☐ **Handles Changes:** Adapts to new challenges easily.
- ☐ **More Success:** Increases the chances of project success.

1.7 Obstacles in Project Management

Here are common obstacles in project management:

- ☐ **Unclear Goals:** Not knowing what to do.
- ☐ **Poor Planning:** Bad cost and time planning.
- ☐ **Limited Resources:** Not enough time, money or people.
- ☐ **Poor Risk Management:** Not identifying and solving problems early.
- ☐ **Poor Time Management:** Tasks not done on schedule (time).
- ☐ **Poor Cost Management:** Overspending or wasting money.
- ☐ **Poor Communication:** Team members not sharing information properly.
- ☐ **Technical Problems:** Tools or software not working.
- ☐ **Poor Skills:** Team members not having the proper knowledge and skills.
- ☐ **Scope Changes:** Adding extra work without proper control.
- ☐ **Team Conflicts:** No mutual understanding between team members.
- ☐ **Tight Deadlines:** Unrealistic time given to complete tasks.
- ☐ **Resistance to Change:** People not wanting to try new things.

1.8 Project Manager and His Role

Project Manager Roles are as follows:

- ☐ **Planning:** Creating detailed project plans, schedules and budgets.
- ☐ **Resource Management:** Using people, money and tools effectively.
- ☐ **Quality Management:** Complete the work with required standards.
- ☐ **Risk Management:** Finding and fixing problems early.
- ☐ **Decision Making:** Choosing the best actions to complete project.

- ☐ **Leading the Team:** Guiding and motivating team members.
- ☐ **Solving Problems:** Fixing issues and resolving conflicts.
- ☐ **Communicating:** Make proper communication between team members and stakeholders.
- ☐ **Track Progress:** Make sure tasks are completed on time and within budget.
- ☐ **Adaptability:** Changing strategies and plan as per the situation.
- ☐ **Working with Stakeholders:** Managing their expectations and needs.
- ☐ **Reporting:** Giving updates on the project progress to the team and stakeholders.

1.9 Project Consultants

- ☐ A Project consultant is an expert person hired to provide expert advice.
- ☐ Specialized knowledge of Project consultant helps manage and complete the project effectively and successfully.
- ☐ A Project consultant assists the Project Manager in planning, problem-solving, cost management, time management and implementing best practices.
- ☐ Project consultants fill the skill gap when there is no expert available in the project team.

Types of Project consultants:

- i. In-house consultant**
- ii. Outside consultant**

i. In-house consultants:

- ☐ A separate department, within the project management organization is called in-house consultants.
- ☐ They are employees within an organization who provide advice and expertise
- ☐ They look after work such as detailed Engineering drawing and preparation of technical specification etc.

ii. Outside consultants:

- ☐ When the jobs cannot be done by the in-house consultants, then the organization may hire an external consultant.
- ☐ They do not work directly within the company but are engaged temporarily to assist project.
- ☐ Outside consultants can be classified as
 - ☐ Indigenous consultant (within the country)
 - ☐ Foreign consultants (outside the country)

Some of the consultancy firms in India are as follows:

- ☐ Wipro Consulting Services
- ☐ Tata Consultancy Services (TCS)
- ☐ Infosys Consulting
- ☐ HCL Technologies
- ☐ IBM Consulting
- ☐ L&T Infotech
- ☐ Bain & Company India
- ☐ KPMG India
- ☐ Accenture India
- ☐ Boston Consulting Group (BCG) India
- ☐ CGI group
- ☐ HP Enterprise Technology Services
- ☐ Kirloskar Consultancy LTD
- ☐ Birla Technical Services
- ☐ Technical Consultancy Organization
- ☐ Science and Technology Entrepreneurship Park Ltd.
- ☐ Dastur and Co Ltd.
- ☐ Power Consultancy Services India Pvt Ltd.



1.10 Meaning of Operation.

- ☐ Operation refers to the day-to-day activities and processes that an organization performs to produce products (goods or services).
- ☐ It focuses on maintaining efficiency, managing resources and ensuring profit.

1.11 Difference between Project and Operation:

Sl #	Project	Operation
1	Temporary	Repetitive
2	Unique	Continuous cycle
3	Create new product, service or process	Product, service or process already created and is in use
4	Performance, cost and time are uncertain	Performance, cost and time are known
5	Developing a new system	System already exists, maintaining and sustaining
6	Unexpected inputs and outputs	Expected inputs and outputs
7	More risk, usually done for the first time	Fewer risk as they are repeated many times
8	When objectives are achieved the project ends	Multiple objectives to be achieved again and again

Examples for Project and Operation:

Project	Operation
• Building a new school.	• Running a school.
• Launching a new smartphone.	• Producing smartphones.
• Constructing a highway.	• Managing highway tolls.
• Setting up a solar power plant.	• Generating electricity through the solar plant.
• Designing a new website.	• Hosting and updating the website.
• Writing a book.	• Publishing books regularly.
• Starting a YouTube channel.	• Creating regular video content.

1.13 Process in Project Management

- ❑ Process in project management refers to a series of steps or activities followed to plan, execute, monitor and complete a project effectively.
- ❑ It is the application of knowledge, skills, tools and techniques to complete project activities.
- ❑ This involves the application of five project management process such as:
 - i. Initiating
 - ii. Planning
 - iii. Executing
 - iv. Monitoring and controlling
 - v. Closing.
- ❑ These processes ensure the project stays on track and meets its objectives.
- ❑ The Project Manager is the person responsible for finishing the project objectives.

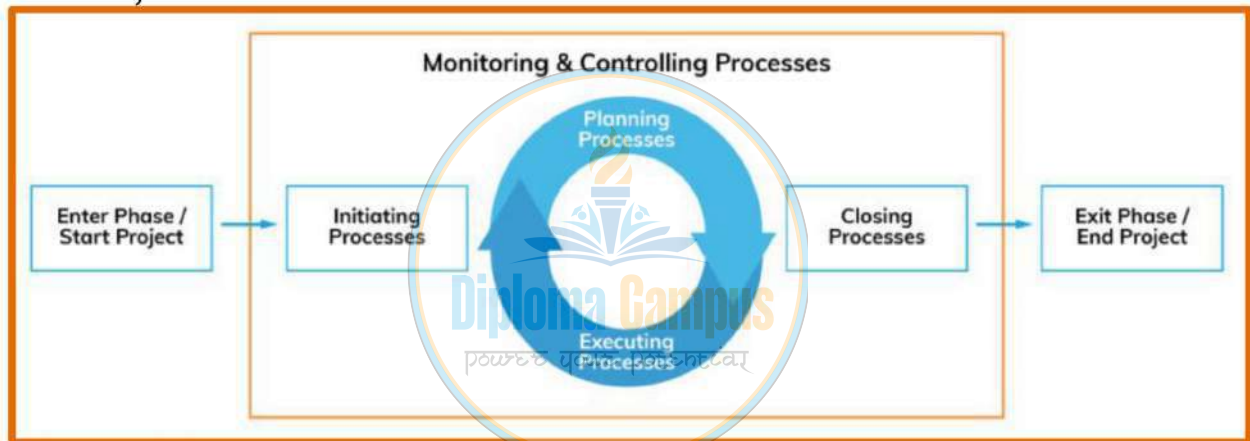


Fig. Process in Project Management

1.14 Project Objectives and Project Scope:

Project Objectives:

- ❑ Project Objectives are specific, measurable outcomes that need to be achieved to meet the overall goal of a project.
- ❑ It defines the desired outcomes and the timeframe required to complete the goal (Big picture goal, the why of the project).
- ❑ Project objectives clarify the purpose of the project and provide a plan to achieve success.
- ❑ Objectives are clear, time-bound and aligned with the goals.

Example for Project Objectives:

For the house construction project, the objectives could be:

- Complete the house in 12 months.
- Build a 3-bedroom house with eco-friendly materials.
- Stay within a budget of ₹40,00,000 (Forty Lakh Rupees)

Project Scope:

- ☐ Project Scope defines the boundaries of a project.
- ☐ It explains what is included and what is excluded.
- ☐ It defines the work to be done, deliverables (output) to be produced and resources required to complete.
- ☐ A well-defined project scope ensures that the team focuses on the goals and avoids unnecessary tasks.

Example for Project Scope:

For the house construction project:

- Designing the house layout, building the structure and installing plumbing and electrical systems are included.
- However, landscaping such as garden, creating pathways, adding decorative features outside the house, a swimming pool and future maintenance such as regular repairs are excluded.

Key Difference between Project Objectives and Project Scope:

Sl #	Project Objectives	Project Scope
1.	Objectives focus on why project is being done.	Scope focuses on what needs to be done.
2.	Objectives are defined in PID (Project Initiation Document)	Scope are defined in WBS (Work Breakdown Structure)
3.	Objectives define measurable outcomes to be achieved to meet overall goals of a project.	Scope defines output, outcomes, benefits and work required.
4.	Objectives define desired outcomes and success criteria.	Scope defines boundaries, what is included and what is excluded.
5.	Objectives focus on “what” we have to do and “why” we have to do.	Scope focus on “how” we can complete.
6.	Objectives are measured by the completion of specified task at given time and fixed cost.	Scope is measured by the completion of specified task.
7.	Objectives are generally fixed.	Scope can be adjusted through change control processes to support necessary modifications during the project lifecycle.
8.	Project Objectives focus on the end goals or results.	Project Scope focus on tasks and deliverables.

